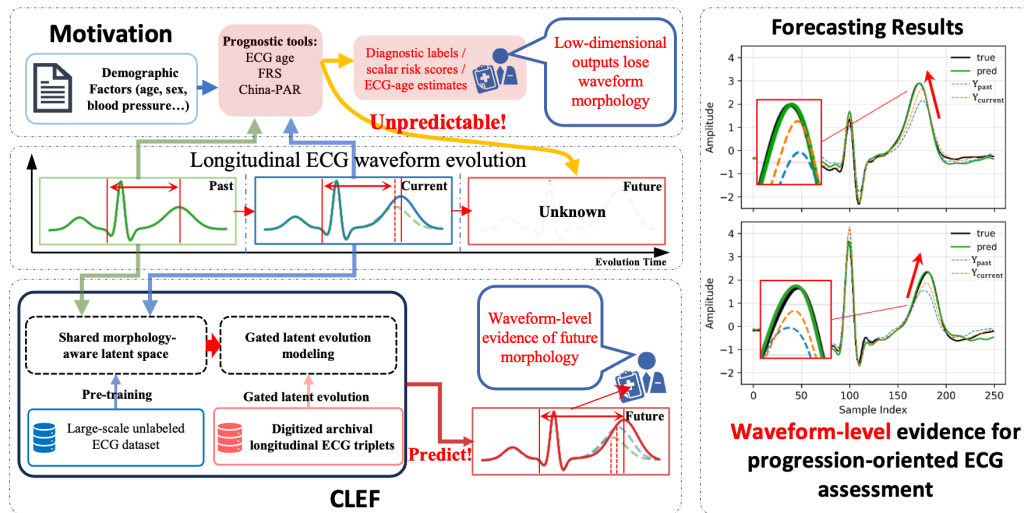


Graphical Abstract

A Cross-Temporal Latent Evolution Framework for Year-Scale ECG Waveform Forecasting from Historical Records

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Highlights

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- Formulates year-scale ECG waveform forecasting beyond descriptor tracking and static prediction.
- Frames future waveform forecasting as shared latent alignment followed by latent evolution.
- Introduces adaptive gated latent evolution to separate identity preservation from progression modeling.
- Demonstrates morphology-preserving year-scale ECG waveform forecasting from heterogeneous archival records.

A Cross-Temporal Latent Evolution Framework for Year-Scale ECG Waveform Forecasting from Historical Records

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Abstract

Cardiovascular diseases progress through silent electrophysiological remodeling, yet existing prognostic tools reduce electrocardiography (ECG) signals to diagnostic labels or scalar risk scores, while ECG waveform forecasting may provide earlier, interpretable waveform-level evidence for cardiac risk assessment. However, subject-specific morphology, inter-subject variability, and observation-domain variation make year-scale progression patterns difficult to capture and isolate. To address these challenges, we propose the Cross-Temporal Latent Evolution Framework (CLEF) for forecasting future ECG waveforms. First, CLEF recovers analyzable digital waveforms from paper-based ECG records to establish a standardized sig-

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nal basis for cross-visit alignment. Second, instead of predicting future ECG waveforms in the raw signal domain, CLEF frames year-scale waveform forecasting as cross-temporal latent evolution: multi-visit ECGs are projected into a shared morphology-preserving latent space learned from unlabeled ECG data, where cross-visit observations are represented in comparable coordinates with reduced sensitivity to observation-domain variation. Finally, CLEF employs gated latent evolution to separate identity-preserving morphology from progression-sensitive changes and infer a future-consistent latent state for waveform reconstruction. Compared with a time-series forecasting baseline, CLEF reduces normalized root mean square error from 0.106 to 0.077 while preserving waveform morphologies. These results support CLEF as a feasible basis for earlier, morphology-based cardiac risk assessment under heterogeneous archival conditions.

Keywords: longitudinal electrocardiography, representation learning, latent evolution modeling, waveform forecasting, paper ECG digitization

1. Introduction

Cardiovascular diseases (CVDs) remain a leading cause of death worldwide and are associated with substantial morbidity and disability [1]. Electrocardiography (ECG) is one of the most established and widely used tools for cardiac assessment [2]. In current clinical practice and computer-aided systems, ECG analysis predominantly follows a snapshot diagnosis paradigm, i.e., identifying cardiac conditions from a single recording [3, 4]. However, snapshot-based diagnosis only provides limited insight into longitudinal electrophysiological evolution, while many cardiovascular abnormalities develop through silent remodeling over time [5, 6]. For instance, longitudinal changes in waveform-derived markers, such as the heart rate-corrected QT (QTc) interval and QRS duration, have been associated with mortality and sudden cardiac death [7, 8]. Clinicians can assess such marker changes only when a subsequent ECG becomes available, typically during periodic follow-up or after clinical symptoms emerge. By estimating patient-specific future morphology before the next follow-up recording is available, year-scale ECG waveform forecasting may therefore provide earlier, interpretable waveform-level evidence for cardiac risk assessment.

Nevertheless, current cardiac risk assessment remains limited in its ability to represent waveform-level progression. Traditional risk models, such as

Framingham and China-PAR, rely mainly on demographic and clinical variables to estimate population-level cardiovascular risk [9, 10]. Longitudinal ECG studies further show that serial changes in predefined descriptors, abnormality patterns, or scalar measurements are associated with subsequent cardiovascular outcomes [11, 12, 13]. Recent deep learning studies have advanced ECG analysis by learning nonlinear representations directly from raw waveforms. For example, deep ECG models can identify patients in normal sinus rhythm who are at elevated risk of future atrial fibrillation [14], and ECG-age studies suggest that learned representations may encode cumulative electrophysiological variation related to cardiac aging and progression [15]. These findings suggest that raw ECG waveforms contain latent information reflecting physiological evolution and potential disease progression beyond visible diagnostic patterns. Yet this information is primarily exploited through diagnostic labels, scalar risk scores, or intra-record signal modeling targets. Consequently, year-scale ECG morphology evolution remains insufficiently modeled at the waveform level.

These limitations motivate a paradigm shift from descriptor-level assessment to future ECG waveform forecasting. This formulation treats year-scale ECG evolution as a waveform-level prediction target rather than reducing it to scalar risk scores, thereby providing a more interpretable basis for earlier progression-oriented cardiac assessment. Yet realizing this shift is not a standard time-series forecasting problem. As physiological time series, ECG waveforms exhibit complex dynamics involving both non-stationarity and non-linearity [16]. However, multi-year ECG evolution reflects slow electrophysiological remodeling rather than short-horizon signal extrapolation. Sequence models, such as recurrent neural networks and Transformer-based forecasting models, usually learn temporal dependencies through shared transition patterns across time steps [17, 18]. They are better suited to dense longitudinal data with many temporally ordered observations per subject. In contrast, historical ECG records often contain only few observations per subject, separated by follow-up intervals that are generally measured in years. This sparsity makes generic sequence fitting unreliable for year-scale waveform forecasting.

Beyond the sparsity of historical records, strong subject-specific ECG patterns and large inter-subject differences make common year-scale progression patterns difficult to capture. A forecasting model therefore has to preserve patient-specific morphology while selectively capturing progression-related changes. Otherwise, it may either preserve source morphology with-

out learning a meaningful progression direction, or generate progression-consistent waveforms that become detached from the source subject. This challenge is further amplified by the heterogeneity of archival ECG records, where observation-domain variation arises from acquisition inconsistency, phase shifts, signal-quality variation, and paper-to-digital conversion artifacts [19, 20]. As a result, year-scale evolution patterns are difficult to isolate from cross-visit waveform differences. Together, these factors limit individualized ECG waveform forecasting from historical ECG records.

To address these challenges, we reform the problem of future ECG waveform forecasting into cross-temporal latent evolution of ECG representations and propose the Cross-Temporal Latent Evolution Framework (CLEF). The framework first recovers analyzable digital waveforms from paper-based archival ECG records and establishes a standardized signal basis for cross-visit alignment. Based on the digitized waveforms, we construct a longitudinal ECG dataset for waveform evolution modeling with 14,304 within-subject triplets across three consecutive annual visits. Instead of directly predicting raw waveforms, CLEF performs future inference in a shared latent space. Specifically, multi-visit ECGs are projected into a morphology-preserving latent space learned from large-scale unlabeled ECG data. This shared representation provides comparable cross-visit coordinates for longitudinal modeling and reduces sensitivity to observation-domain variation. CLEF then uses gated latent evolution to adaptively separate identity-preserving information from progression-sensitive evolution. The identity-preserving component retains patient-specific morphology, while the progression-sensitive component models longitudinal change. The resulting future-consistent latent state is decoded into a future ECG waveform rather than a predefined descriptor vector. With these efforts, CLEF can accurately predict the third-year waveform given two consecutive annual ECG observations, by evolving the corresponding latent representation. In longitudinal ECG forecasting, CLEF reduces the normalized root mean square error from 0.106 for the LSTM baseline to 0.077, while preserving key waveform morphologies. These results suggest that CLEF provides a feasible framework for waveform-level ECG forecasting from heterogeneous archival records and may support earlier, interpretable, morphology-based evidence for cardiac risk assessment.

Our contributions are summarized as follows:

- We formulate year-scale ECG waveform forecasting from heterogeneous historical records as a cross-temporal progression modeling task, mov-

ing beyond descriptor tracking and static prediction toward waveform-level longitudinal assessment.

- We propose CLEF, a cross-temporal latent evolution framework that establishes a standardized signal basis, aligns cross-visit ECGs in a shared morphology-preserving latent space, and models future progression through adaptive gated latent evolution.
- We construct a longitudinal ECG dataset with 14,304 within-subject triplets from digitized archival records and demonstrate that CLEF improves year-scale waveform forecasting while preserving key future ECG morphologies.

The rest of this paper is organized as follows. Section 2 reviews related work, Section 3 presents the proposed method, Section 4 reports the experimental results, Section 5 discusses the findings and limitations, and Section 6 concludes the paper.

2. Related Work

This study lies at the intersection of three research areas: ECG representation learning, longitudinal ECG analysis, and self-supervised learning. Across these areas, the central unresolved problem is how to learn ECG representations that preserve morphological information, remain comparable across visits, and support longitudinal modeling under heterogeneous conditions.

2.1. Descriptor-based and learned ECG representation

Early computational ECG analysis primarily followed the logic of clinical interpretation by focusing on predefined waveform components, such as the P wave, QRS complex, T wave, rhythm patterns, and interval measurements [21]. Based on these handcrafted descriptors, conventional machine learning models, including linear discriminants [22] and support vector machines [23], were developed for diagnostic classification. While these approaches established an early framework for computer-aided ECG analysis, they depend heavily on manual feature engineering and are highly sensitive to noise, morphological variability, and acquisition inconsistencies in real-world data [24].

Handcrafted ECG descriptors remain widely used because their explicit definitions enable direct comparison across recordings and visits [25]. However, such comparability is obtained by reducing each ECG to a limited set of low-dimensional measurements. While these representations are suitable for rule-based interpretation, interval analysis, and scalar risk estimation, they are not designed to preserve the full morphological structure of ECG waveforms or to support waveform-level forecasting.

Recent deep learning methodologies have substantially advanced ECG analysis by learning hierarchical representations directly from raw waveforms. Convolutional neural networks (CNNs) [26], recurrent neural networks (RNNs) [27], long short-term memory (LSTM) networks [28], and gated recurrent unit (GRU) models [29] have demonstrated strong performance in automated diagnosis and prognostic prediction. Furthermore, related studies have investigated latent biomarkers, such as ECG-age [30], indicating that learned representations can encode information beyond conventional handcrafted parameters. Nevertheless, most existing deep ECG models still operate on single-time-point recordings, targeting discrete class labels, risk scores, or surrogate indices. Although these models improve expressive power, they do not directly address the challenge of learning high-dimensional ECG representations that remain comparable across visits for longitudinal waveform-level analysis.

2.2. Longitudinal ECG analysis and future waveform forecasting

Cardiac electrophysiology evolves over time, and repeated ECG recordings from the same individual therefore contain information that is not available from isolated snapshot analysis. Previous studies have demonstrated that serial ECG changes provide prognostic information beyond that available from a single recording [31, 32], further supporting the value of multi-visit ECG analysis. However, most existing longitudinal ECG studies focus primarily on risk stratification, disease progression assessment, or clinical event prediction, rather than direct forecasting of future ECG waveform morphology.

This distinction is methodologically important because the two tasks require different modeling assumptions. Predicting a future label or risk score requires mapping current observations to a predefined outcome, whereas future waveform forecasting requires preserving high-dimensional morphology together with an explicit model of temporal transition. Early studies of

ECG dynamics relied mainly on traditional time-series methods, such as autoregressive integrated moving average (ARIMA) models [33]. More recent neural approaches have explored signal generation, short-horizon forecasting, and synthetic ECG generation [34, 35, 36]. Other waveform-oriented studies have remained focused primarily on intra-record tasks, such as reconstruction, denoising, imputation, or anomaly detection [37, 38, 39]. Despite their contributions, these studies do not address year-scale forecasting from sparse observations acquired across longitudinal follow-up visits.

A further challenge is that ECGs acquired at different visits are affected by acquisition inconsistency, temporal phase shifts, and storage heterogeneity. This challenge becomes more pronounced in retrospective cohorts, where a substantial portion of historical ECG data may exist only as paper print-outs or scanned ECG reports rather than as standardized digital signals. Consequently, longitudinal ECG forecasting requires not only temporal prediction but also the learning of robust representations under heterogeneous conditions before temporal evolution can be modeled reliably.

2.3. Self-supervised ECG representation learning for cross-temporal modeling

Large-scale, high-quality labeled ECG datasets remain expensive and difficult to acquire, a limitation that is even more pronounced for longitudinal ECG data, which require consistent follow-up and annotation. Consequently, self-supervised and unsupervised learning have emerged as effective strategies for leveraging large-scale unlabeled ECG recordings to learn generalizable signal representations [40, 41]. Among these approaches, masked autoencoders (MAEs) [42] are particularly relevant in this context. By reconstructing masked input segments, MAEs encourage the encoder to learn both local morphological structure and global waveform organization. This property is crucial for future ECG waveform forecasting, which requires a shared latent space that preserves morphology while supporting temporal evolution modeling and waveform reconstruction. Compared with purely discriminative pretraining, reconstruction-based pretraining is better aligned with this objective because it preserves signal structures required for subsequent latent-state evolution and waveform reconstruction.

When pretrained on large public repositories, MAE-based models provide a robust foundation for learning morphology-aware latent spaces [43, 44]. Existing studies have primarily applied these representations to downstream tasks such as classification, abnormality detection, and within-record reconstruction. While these works demonstrate the effectiveness of large-scale

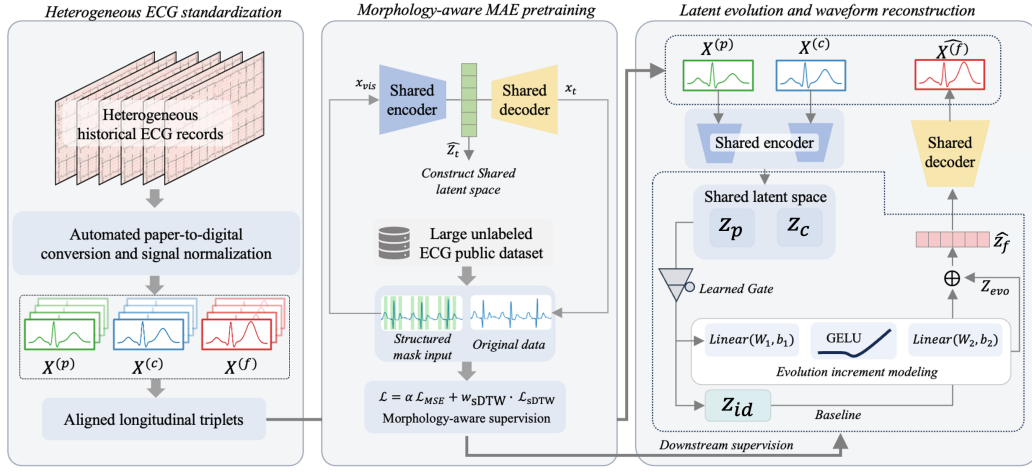


Figure 1: Overview of CLEF for year-scale ECG waveform forecasting from heterogeneous historical records. First, historical ECG records from past and current visits are normalized into a shared signal domain. Then, a pretrained masked autoencoder (MAE) encodes these cross-visit ECG segments into a shared latent space. Within this space, a latent evolution module infers a future-consistent latent state by combining identity-preserving morphology with progression-sensitive cross-visit evolution. Finally, the inferred latent state is decoded into the future ECG waveform.

pretraining, they generally assume that individual samples can be analyzed independently. This assumption is invalid for longitudinal forecasting, where representations must remain comparable across visits and robust under heterogeneous observation conditions. Furthermore, forecasting requires a latent space that supports temporal evolution rather than only static discrimination. By representing ECGs in a rich latent space instead of through low-dimensional descriptors, longitudinal analysis can be extended toward higher-dimensional prediction targets. From this perspective, future waveform forecasting can be viewed as a higher-dimensional longitudinal objective than scalar risk estimation, because it aims to preserve evolving waveform morphology rather than compress longitudinal progression into a single scalar score.

3. Methods

3.1. Overview

Fig. 1 illustrates the overall CLEF pipeline for future ECG waveform forecasting from heterogeneous historical records. The framework is organized

into three coupled stages: heterogeneous ECG standardization, morphology-aware MAE pretraining, and latent evolution with waveform reconstruction.

First, heterogeneous historical ECG records are converted into analyzable digital signals through automated paper-to-digital conversion and signal normalization. The processed records are then organized into aligned longitudinal triplets, consisting of past, current, and future ECG segments from the same subject. This stage reduces observation-domain heterogeneity caused by report format variation, acquisition inconsistency, temporal phase shifts, and digitization artifacts.

Second, CLEF learns a shared latent representation through morphology-aware masked autoencoder pretraining on large-scale unlabeled ECG data. The shared encoder maps ECG segments into latent representations, while the shared decoder reconstructs waveform morphology from the latent space. This pretraining stage provides a common representation basis in which multi-visit ECGs can be aligned and modeled more reliably than in the raw waveform domain.

Third, the past and current ECG segments are encoded into the shared latent space for downstream forecasting. Within this space, CLEF uses a learned gate to preserve identity-related morphology, while a nonlinear evolution branch models progression-sensitive latent changes. The identity-preserving and progression-sensitive components are combined to infer a future-consistent latent state, which is then decoded into the predicted future ECG waveform. This design supports identity-preserving latent evolution by reducing interference between stable identity-related morphology and subtle cross-visit electrophysiological progression.

3.2. Datasets and preprocessing

The proposed framework is developed using two datasets with distinct purposes. The longitudinal forecasting dataset is constructed from historical ECG reports for downstream model training and evaluation. The unlabeled digital ECG dataset is used for self-supervised pretraining to learn a shared latent space.

3.2.1. Longitudinal forecasting dataset

The longitudinal forecasting dataset was constructed from historical ECG reports, many of which were available only as paper printouts or scanned records. These archival records vary in report format, signal quality, acquisition condition, temporal phase, and digitization artifacts, so cross-visit dif-

ferences may reflect both physiological progression and observation-domain variation.

Paper ECG reports were converted into analyzable one-dimensional signals using our previously developed digitization method [45]. Records with failed digitization, incomplete waveform extraction, unresolved lead overlap, invalid amplitude calibration, or severe residual distortion were excluded.

The final dataset contains 14,304 approximately annual within-subject ECG triplets from 20,591 historical ECG records. Each triplet consists of one past, one current, and one future segment, denoted as $(\mathbf{X}^{(p)}, \mathbf{X}^{(c)}, \mathbf{X}^{(f)})$, with consecutive visits separated by 12 ± 2 months. For subjects with more than three annual records, at most three valid triplets were retained. The dataset was split at the subject level to prevent leakage of subject-specific longitudinal information.

Ethical approval for collecting and using the private longitudinal dataset in this study was granted by the Ethics Committee of Fuwai Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College (ethical approval reference number: 2025-2786).

3.2.2. *Unlabeled pretraining dataset*

The PTB-XL database contains 21,799 ten-second 12-lead ECG recordings from 18,869 patients [46]. Lead II was extracted and resampled from 500 Hz to 300 Hz, yielding 333,208 unlabeled segments for self-supervised pretraining.

3.2.3. *Data preprocessing*

All signals were denoised using a fourth-order Butterworth band-pass filter. For both datasets, each 3-second Lead II segment was aligned by its R-peak to minimize short-term phase variability and position the dominant heartbeat morphology within a consistent temporal neighborhood. The aligned segments were then standardized using Z-score normalization. Both paper-derived ECGs and native digital ECGs from PTB-XL were processed under the same preprocessing protocol, while residual discrepancies in signal quality are explicitly evaluated in the subsequent signal-fidelity stratification analysis.

3.3. *Morphology-aware MAE Pretraining*

A Transformer-based masked autoencoder (MAE) is pretrained on the unlabeled ECG dataset to construct a shared morphology-aware latent space.

Reconstruction-based pretraining is used because the downstream task requires latent representations that preserve waveform structure and support subsequent waveform reconstruction.

Each 3-second ECG segment is partitioned into 36 non-overlapping patches, with each patch comprising 25 samples. These patches are linearly embedded and augmented with positional encodings. The encoder consists of 16 Transformer blocks with multi-head self-attention. During pretraining, 75% of input patches are masked.

Rather than using purely random masking, the pretraining strategy preserves a contiguous visible region spanning a complete P–QRS–T complex. Because all input segments are aligned by the R peak during preprocessing, this visible region is defined as a fixed local window centered on the dominant heartbeat. This structured masking strategy allows the model to observe a physiologically meaningful local event while reconstructing the global waveform organization from incomplete input.

Let $\mathbf{X} = [x_1, \dots, x_N]$ denote the patch sequence, and let

$$\mathbf{X}_{\text{vis}} = \{x_i \mid i \in \mathcal{S}_{\text{PQRST}}\} \quad (1)$$

represent the visible subset corresponding to the preserved P–QRS–T region. The encoder maps this visible subset to a latent representation,

$$\mathbf{F} = E_\phi(\mathbf{X}_{\text{vis}}), \quad (2)$$

and a symmetric Transformer decoder reconstructs the full ECG segment from the latent representation and mask tokens. After pretraining, the encoder and decoder define the shared latent space and waveform reconstruction mapping used in downstream longitudinal forecasting.

3.4. Cross-temporal latent evolution modeling

Given a past segment $\mathbf{X}^{(p)}$ and a current segment $\mathbf{X}^{(c)}$, the pretrained encoder maps these inputs into the shared latent space as

$$\mathbf{z}_p = E_\phi(\mathbf{X}^{(p)}), \quad \mathbf{z}_c = E_\phi(\mathbf{X}^{(c)}). \quad (3)$$

In all experiments, each latent variable is represented as a 128-dimensional vector. Because the same pretrained encoder is used for both visits, the resulting latent variables reside in a shared reference space rather than in visit-specific feature coordinates. This shared space supports longitudinal

forecasting because future inference is performed through latent evolution in comparable coordinates rather than through direct comparison in the raw waveform domain.

Within this shared space, ECG representations still exhibit identity–evolution coupling: relatively stable patient-specific morphology is intertwined with subtle cross-visit electrophysiological progression. CLEF addresses this coupling by decomposing the latent representation into an identity-preserving component and a progression-sensitive evolution component:

$$\hat{\mathbf{z}}_f = \mathbf{z}_{\text{id}} + \mathbf{z}_{\text{evo}}, \quad (4)$$

where $\mathbf{z}_{\text{id}}$ preserves identity-related morphology from the current visit, and $\mathbf{z}_{\text{evo}}$ captures progression-sensitive latent changes inferred from temporally ordered observations.

Direct extrapolation from concatenated features can become unstable when applied to long-term heterogeneous data, because stable subject-specific morphology and evolving longitudinal changes may be mixed within the same transformation. To mitigate this issue, the proposed gated evolution operator introduces an explicit structural inductive bias, inspired by the principles of temporal gating and disentangled representation learning [47, 48].

Specifically, a gate vector is estimated from the concatenated past and current latent states:

$$\mathbf{m}_c = \sigma(\mathbf{W}_g[\mathbf{z}_p; \mathbf{z}_c] + \mathbf{b}_g), \quad (5)$$

where $\mathbf{m}_c \in [0, 1]^d$ controls the adaptive decomposition of the latent space into identity-preserving and progression-sensitive dimensions.

The gate vector $\mathbf{m}_c$ serves as a learned selector that controls how much of the current latent representation is preserved in the future state [49]. By dynamically evaluating the temporal consistency between $\mathbf{z}_p$ and $\mathbf{z}_c$, $\mathbf{m}_c$ selectively identifies latent dimensions associated with identity-related morphology, which are then passed to the future state through $\mathbf{z}_{\text{id}}$:

$$\mathbf{z}_{\text{id}} = \mathbf{m}_c \odot \mathbf{z}_c, \quad (6)$$

where $\odot$ denotes element-wise multiplication.

The evolution branch first estimates an evolution proposal from the temporally ordered latent observations:

$$\tilde{\mathbf{z}}_{\text{evo}} = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1[\mathbf{z}_p; \mathbf{z}_c] + \mathbf{b}_1) + \mathbf{b}_2. \quad (7)$$

In contrast, the complementary gate selects progression-sensitive dimensions from this nonlinear proposal:

$$\mathbf{z}_{\text{evo}} = (1 - \mathbf{m}_c) \odot \tilde{\mathbf{z}}_{\text{evo}}. \quad (8)$$

This gated nonlinear evolution component is designed to capture residual morphology changes that may include progressive ST-segment or T-wave variations.

Together, the identity-preserving component and the progression-sensitive evolution component form the future-consistent latent state:

$$\hat{\mathbf{z}}_f = \mathbf{z}_{\text{id}} + \mathbf{z}_{\text{evo}}. \quad (9)$$

This decomposition separates identity preservation from progression modeling within a shared latent space and reduces the interference between stable subject-specific ECG structure and subtle cross-visit electrophysiological evolution.

3.5. Latent evolution supervision and waveform reconstruction

The inferred latent state $\hat{\mathbf{z}}_f$ represents the predicted electrophysiological state at the next visit. Under the gated latent evolution formulation, this state comprises an identity-preserving component derived from the current visit and a progression-sensitive evolution component inferred from temporally ordered latent observations. To derive an observable ECG signal from this evolved latent representation, the pretrained decoder maps the future-consistent latent state back into the waveform domain:

$$\hat{\mathbf{X}}^{(f)} = D_\psi(\hat{\mathbf{z}}_f). \quad (10)$$

This formulation shows that future waveform reconstruction is achieved by decoding a latent state composed of identity-related morphology and cross-visit electrophysiological evolution.

The training objective supervises this evolution-to-waveform mapping by enforcing both pointwise fidelity and morphology-aware alignment:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + w_{\text{sDTW}} \mathcal{L}_{\text{soft-DTW}}, \quad (11)$$

where $\mathcal{L}_{\text{MSE}}$ constrains amplitude-wise reconstruction fidelity, and $\mathcal{L}_{\text{soft-DTW}}$ promotes temporal alignment and morphological consistency between the reconstructed waveform and the target future waveform [50]. While the decoder

maps the evolved latent state back to the waveform domain, optimization is primarily directed toward the gated latent evolution module to infer a future-consistent state.

During the downstream forecasting training phase, both the pretrained encoder and decoder are initialized using weights from the MAE pretraining stage. The encoder remains frozen to preserve a stable cross-visit latent reference space, whereas the latent evolution module is optimized for future-state inference. Similarly, the decoder is kept frozen and functions as a waveform reconstruction operator that maps the shared latent representation back to the signal domain.

To stabilize the evolution-to-waveform reconstruction process, the contribution of soft-DTW is introduced progressively via a clipped linear warm-up schedule:

$$w_{\text{sDTW}} = \text{clip}_{[0,1]} \left(\frac{\text{epoch} - e_0}{\tau} \right), \quad e_0 = 5, \tau = n_{\text{epoch}} - e_0, \quad (12)$$

where $\text{clip}_{[0,1]}(\cdot)$ denotes truncation to the interval $[0, 1]$. This schedule maintains morphology-aware alignment in an inactive state during the initial training phase, gradually increases its influence following epoch e_0 , and ultimately saturates at 1.0 in epoch n_{epoch} . Consequently, the model is initially guided toward stable waveform-domain reconstruction and then increasingly constrained to preserve temporally aligned morphological structures.

3.6. Training protocol and evaluation metrics

The MAE pretraining phase uses the AdamW optimizer with a learning rate of 1×10^{-4} , a weight decay of 0.05, a batch size of 256, and a training duration of 100 epochs. The downstream CLEF forecasting model also uses AdamW, with a learning rate of 5×10^{-5} , a batch size of 64, and a training duration of 50 epochs. During downstream training, only the latent evolution module is optimized, while the pretrained encoder and decoder remain frozen. Model development and evaluation are implemented using PyTorch on a single NVIDIA RTX A6000 GPU with 48 GB of memory.

For each cross-validation fold, 90% of the triplets are allocated for model development, and the remaining 10% are held out for evaluation. Subject-level separation is strictly maintained across all folds, so all triplets from the same subject are assigned to the same split. Forecasting performance is quantitatively assessed using three complementary metrics: normalized root

mean square error (nRMSE), Pearson correlation coefficient (PCC), and soft-DTW. Lower nRMSE and soft-DTW indicate better performance, whereas higher PCC indicates better waveform agreement.

4. Results

4.1. Overall Forecasting Performance and Qualitative Assessment

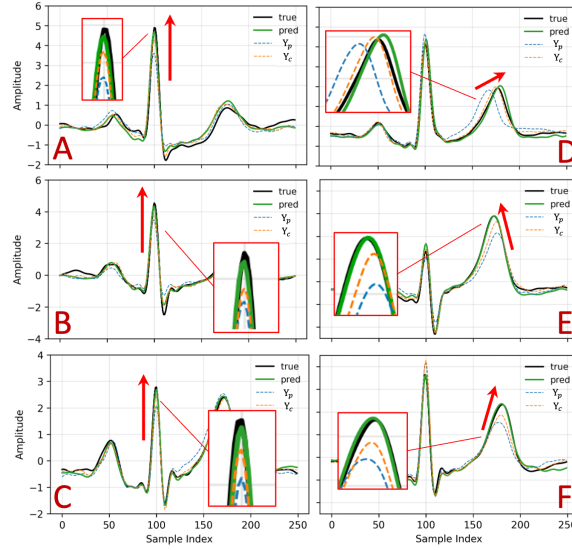


Figure 2: Qualitative examples of future ECG waveform forecasting. Six representative test samples are shown. In each panel, the ground-truth future waveform (true, black) is shown together with the predicted waveform (pred, green) and the two preceding longitudinal waveforms (Y_p , blue dashed line; Y_c , orange dashed line).

Under 10-fold cross-validation, CLEF achieved stable performance for future ECG waveform forecasting. On Z-score-normalized signals, the model achieved an nRMSE of 0.077 ± 0.0006 , a PCC of 0.850 ± 0.0025 , and a soft-DTW distance of 71.44 ± 1.94 . These results indicate that the predicted future waveform remains close to the ground-truth future waveform in both pointwise fidelity and morphology-aware similarity.

These findings demonstrate the feasibility of year-scale ECG waveform forecasting from historical records under rigorous signal normalization and cross-temporal representation alignment. Nevertheless, this task remains fundamentally challenging, as it inherently couples progressive electrophysiology-

ical evolution with substantial cross-visit heterogeneity in the observation domain.

Fig. 2 provides qualitative examples of future ECG waveform forecasting. The predicted waveforms are not simple copies of either the past or current observations. Instead, CLEF produces future waveforms that follow morphologically interpretable changes and remain close to the true future morphology. Representative examples show variation in R-wave amplitude and repolarization patterns, such as T-wave amplification or apparent QT prolongation. These examples further support the view that CLEF captures longitudinal morphological evolution rather than trivially propagating the most recent waveform pattern.

4.2. Comparison with heuristic, linear, and neural forecasting baselines

Table 1: Ten-fold cross-validation results for future ECG waveform forecasting. Lower is better for nRMSE and soft-DTW; higher is better for PCC.

Model	nRMSE↓	PCC↑	soft-DTW↓
<i>Heuristic baselines</i>			
Year1 Pers.	0.134 ± 0.001	0.566 ± 0.006	189.3 ± 6.2
Year2 Pers.	0.132 ± 0.001	0.573 ± 0.006	191.3 ± 4.3
Year Mean	0.114 ± 0.001	0.653 ± 0.005	216.8 ± 3.7
Lin. Extra.	0.243 ± 0.002	0.361 ± 0.007	992.7 ± 15.3
<i>Linear baselines</i>			
Ridge	0.105 ± 0.001	0.700 ± 0.003	293.9 ± 3.2
PLS	0.110 ± 0.000	0.667 ± 0.004	391.3 ± 3.2
PCA+Ridge	0.104 ± 0.001	0.705 ± 0.004	307.9 ± 3.2
<i>Neural baselines</i>			
MLP	0.098 ± 0.001	0.792 ± 0.003	200.2 ± 5.4
Gated Evo-MLP	0.096 ± 0.001	0.799 ± 0.006	195.7 ± 8.4
LSTM	0.106 ± 0.001	0.687 ± 0.005	287.4 ± 6.3
CLEF (ours)	0.0770 ± 0.0006	0.8500 ± 0.0025	71.44 ± 1.94

Abbreviations: Year1 Pers., directly copies the first-year ECG; Year2 Pers., directly copies the second-year ECG; Year Mean, point-wise average of years 1 and 2; Lin. Extra., linear extrapolation with $\hat{\mathbf{x}}_3 = 2\mathbf{x}_2 - \mathbf{x}_1$; Ridge, multi-output linear regression with ℓ_2 regularization; PLS, partial least squares regression in a shared latent space; PCA+Ridge, PCA followed by ridge regression; MLP, plain feed-forward network on concatenated two-year ECGs; Gated Evo-MLP, MLP latent evolution baseline with a gated increment; LSTM, recurrent baseline with two-channel temporal input; CLEF, the proposed method.

To evaluate the proposed framework comprehensively, we compared CLEF with a diverse set of heuristic, linear, and neural forecasting baselines. These

baselines were selected to represent progressively stronger alternatives, from direct temporal continuation to conventional statistical regression and lightweight neural predictors. This comparison allows us to assess both whether future ECG waveform forecasting is feasible in the evaluated setting and which modeling components are most critical for strong performance under heterogeneous longitudinal conditions. All baselines were trained and evaluated on the same subject-level folds using the same preprocessing, normalization, and evaluation protocol as CLEF. The results are summarized in Table 1.

As shown in Table 1, CLEF achieves the best performance across all three metrics and outperforms heuristic, linear, and neural alternatives by a substantial margin. In particular, CLEF achieves an nRMSE of 0.0770 ± 0.0006 , a PCC of 0.8500 ± 0.0025 , and a soft-DTW of 71.44 ± 1.94 . These results show superior pointwise fidelity, higher waveform correlation, and substantially better morphology-aware alignment.

The heuristic baselines perform worst overall. Both persistence-based baselines and simple averaging yield limited predictive quality, while direct linear extrapolation performs poorly across all metrics. These results indicate that year-scale ECG waveform evolution cannot be adequately captured by naive temporal continuation or simple signal-level combination alone.

The linear baselines improve substantially over the heuristic rules. This improvement suggests that paired historical ECG observations contain predictive structure that can be exploited by learned mappings. However, their performance remains substantially below that of CLEF, especially in morphology-aware alignment. This pattern indicates that conventional statistical regression and low-dimensional linear projection remain insufficient to capture morphology-consistent longitudinal evolution under heterogeneous follow-up conditions.

The neural baselines further improve performance. This improvement suggests that nonlinear modeling is beneficial for this task. However, even the strongest neural baseline remains notably below CLEF across all metrics. This remaining gap suggests that the advantage of CLEF does not arise solely from the use of a neural predictor, but from the combination of shared representation learning, explicit latent evolution modeling, and waveform-domain reconstruction.

Taken together, these comparisons provide consistent evidence across three levels of baselines. The heuristic results show that simple continuation is inadequate, the linear results show that conventional statistical mapping remains insufficient, and the neural results show that generic nonlinear pre-

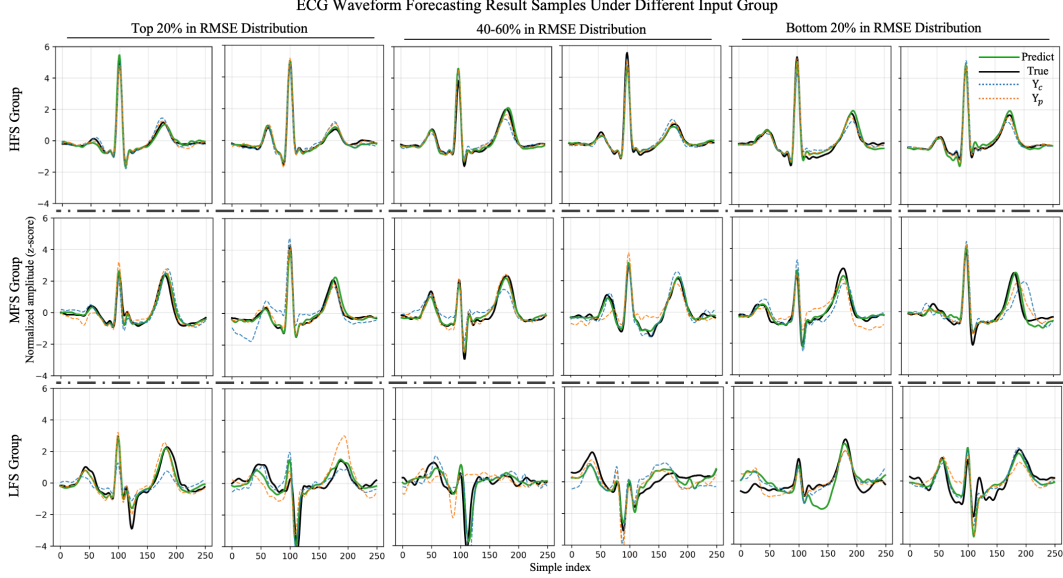


Figure 3: Representative examples of ECG waveform forecasting. Eighteen test samples are shown, grouped by input signal quality: High-Fidelity Signals (HFS), Moderate-Fidelity Signals (MFS), and Low-Fidelity Signals (LFS). Each plot displays the past ECG (Y_p), current ECG (Y_c), predicted future waveform, and the ground truth.

diction alone does not fully resolve the problem. Accurate year-scale ECG waveform forecasting therefore benefits from explicitly organizing cross-visit observations in a shared latent space and modeling their evolution before waveform reconstruction.

4.3. Performance under different signal-fidelity regimes

To evaluate the robustness of the forecasting model under different signal quality levels, the dataset was stratified into high-fidelity signals (HFS), moderate-fidelity signals (MFS), and low-fidelity signals (LFS). These three groups account for approximately 80%, 10%, and 10% of the longitudinal dataset, respectively. This stratification follows the quantitative ECG quality assessment method proposed by Zhao et al. [51], in which the corresponding score ranges are mapped to the quality categories of “Excellent”, “Barely acceptable”, and “Unacceptable”.

Fig. 3 illustrates representative examples of ECG waveform forecasting across input-quality groups. The top row shows cases in which both the past (Y_p) and current (Y_c) input signals are classified as HFS. Across different

Table 2: Performance of CLEF under different signal-fidelity groups (mean $\pm$ std).

Group	nRMSE	PCC	soft-DTW
HFS	0.073 ± 0.022	0.850 ± 0.080	53.17 ± 24.05
MFS	0.077 ± 0.025	0.827 ± 0.113	55.85 ± 26.14
LFS	0.093 ± 0.032	0.795 ± 0.120	77.33 ± 25.11

nRMSE ranges, the model uses these high-quality inputs to generate a forecasted waveform (green line) that remains close to the ground-truth waveform (black line). The middle row shows cases in which at least one input signal is classified as MFS. Although these inputs exhibit noticeable amplitude fluctuations, the predicted waveforms generally preserve the main morphological structure of the true ECG, including key peaks and intervals, across different nRMSE ranges. The bottom row highlights the most challenging cases, in which both input signals (Y_p and Y_c) are classified as LFS due to severe noise artifacts or pronounced morphological abnormalities. Although prediction error is higher in these cases, the model still generates morphologically coherent waveforms and retains the major temporal and amplitude characteristics of the target waveform, avoiding unrealistic or unstable outputs.

As reported in Table 2, performance decreases as signal fidelity declines, but the degradation remains moderate across signal-fidelity regimes. Relative to HFS, the MFS group shows only a modest increase in nRMSE from 0.073 to 0.077 and soft-DTW from 53.17 to 55.85, whereas the LFS group shows a more pronounced decline across all metrics, with nRMSE increasing to 0.093, PCC decreasing to 0.795, and soft-DTW increasing to 77.33. Even so, the LFS group still shows moderate waveform agreement. This result suggests that CLEF preserves the overall waveform structure under more challenging observation conditions. Overall, although archival signal quality affects forecasting accuracy, CLEF retains meaningful waveform agreement across different fidelity regimes.

4.4. Preservation of diagnostically relevant waveform structure

To evaluate whether the predicted waveforms preserve diagnostically relevant ECG structures, we analyzed key parameters derived from the PQRST complex, which serve as the foundation for standard ECG interpretation. Accurate quantification of these parameters, including wave amplitudes, in-

tervals, and durations, requires precise detection of the P, Q, R, S, and T peaks. Therefore, we first assessed peak detection accuracy in the predicted waveforms.

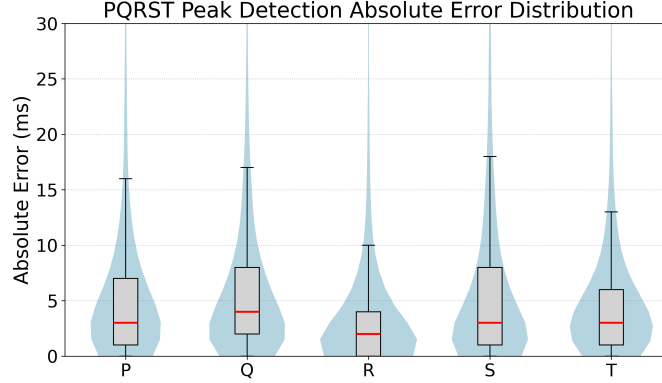


Figure 4: Distribution of absolute timing error for PQRST peak detection. Violin plots illustrate the distribution of absolute timing errors (in milliseconds) for automated detection of P, Q, R, S, and T wave peaks in the predicted ECG waveforms. Each plot combines a boxplot (showing median, interquartile range, and whiskers) with a kernel density estimation of the error distribution.

The detection of these critical peaks was highly accurate, with median absolute errors remaining below 10 ms across all peaks. Notably, the R-wave showed the lowest median error (Fig. 4). Based on these results, eight clinically relevant waveform features were further analyzed: P-, R-, and T-wave amplitudes, PR interval, QRS duration, P-wave duration, T-wave duration, and QT interval. These features cover the major electrophysiological phases of a cardiac cycle: (1) P-wave amplitude and duration characterize atrial depolarization and may indicate left atrial enlargement or intra-atrial conduction delay [52, 53, 54]; (2) PR interval reflects atrioventricular conduction time, with prolongation potentially indicating AV block [55]; (3) R-wave amplitude and QRS duration characterize ventricular depolarization, with abnormalities associated with ventricular hypertrophy, delayed conduction, bundle branch block, or ventricular rhythm [56, 57, 58]; and (4) T-wave amplitude, T-wave duration, and QT interval reflect ventricular repolarization and total ventricular electrical activity, with abnormalities associated with ischemia, electrolyte imbalance, repolarization disturbance, and arrhythmic risk [59, 60, 61, 62].

Statistical analysis was then performed to compare the predicted values

against the ground truth (Fig. 5), with correlation and agreement assessed using scatter plots and Bland-Altman plots, respectively.

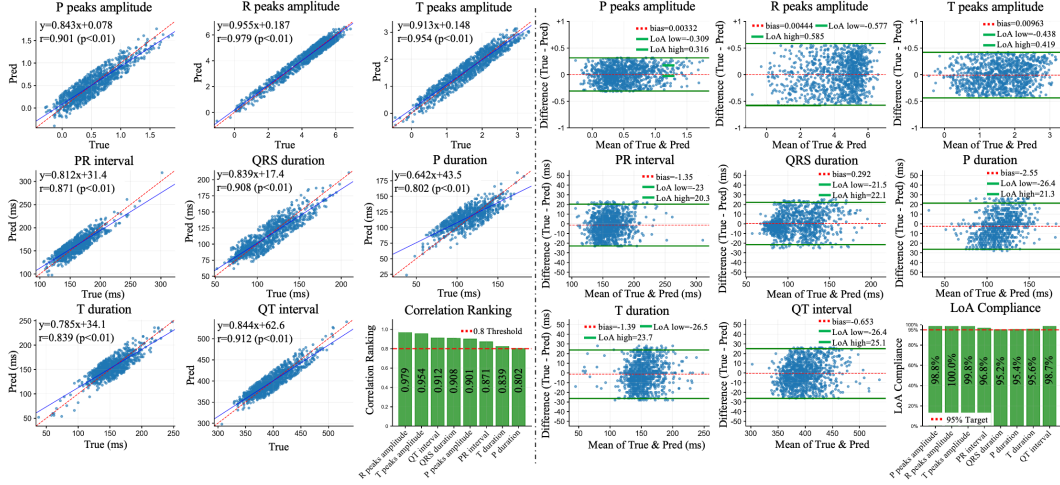


Figure 5: Statistical agreement analysis of ECG parameter estimation. Left panels: scatter plots showing the correlation between predicted (Pred) and true (True) values. Right panels: Bland-Altman plots assessing the agreement between predicted and true values. Bottom row, center and right: summary bar charts, with one ranking the parameters by their PCC and the other showing the percentage of data points within the 95% LoA.

The left panels show the relationship between the predicted and true values for each metric. Across all eight parameters, the data points are tightly clustered around the identity line ($y = x$). This distribution indicates a strong positive linear relationship. The high PCC values further support this observation, with the R-peak amplitude showing the highest correlation (PCC = 0.979), followed by the T-peak amplitude (PCC = 0.954). Notably, even the parameter with the lowest correlation, P-wave duration, still showed a strong correlation (PCC = 0.802). The “Correlation Ranking” bar chart further shows that all analyzed ECG features have high correlation coefficients, with five of the eight parameters achieving a PCC greater than 0.90. Bland-Altman plots [63] (right panels for each metric) were used to assess agreement by visualizing the differences between the two measurements relative to their means. Systematic bias was low across all parameters, with the mean difference (solid red line) remaining close to zero. Furthermore, the “LoA” chart confirms that, for every parameter, a large proportion of samples fall within the 95% limits of agreement.

4.5. Ablation of latent evolution modeling and waveform reconstruction supervision

To evaluate the individual contributions of gated nonlinear latent evolution and morphology-aware waveform reconstruction supervision, the full CLEF was compared with two reduced variants. The first variant removes the soft-DTW term from the waveform reconstruction objective (w/o sDTW). The second variant replaces the gated nonlinear latent evolution module with a linear latent extrapolation baseline (LLF), in which the future latent state is inferred through a strictly linear combination of temporally ordered latent observations within the same shared latent space. The results are summarized in Table 3.

Table 3: Ablation results for latent evolution modeling and waveform reconstruction supervision (mean $\pm$ std).

Model	nRMSE	PCC	soft-DTW
CLEF	0.077 ± 0.0006	0.850 ± 0.0025	71.44 ± 1.94
w/o sDTW	0.076 ± 0.0005	0.855 ± 0.0020	98.88 ± 1.92
LLF	0.091 ± 0.0009	0.797 ± 0.0043	87.42 ± 2.74

Replacing the gated nonlinear latent evolution module with the linear latent extrapolation (LLF) baseline led to a substantial decline across all performance metrics. This degradation is consistent with the proposed design rationale: strictly linear evolution within the shared latent space is insufficient to preserve relatively stable patient-specific morphology while modeling longitudinal progression. Without the gating mechanism, linear extrapolation appears more vulnerable to feature entanglement across visits. As a result, it tends to mix stable identity-related morphology with cross-visit progression-sensitive changes and produces less accurate inference of the future latent state.

In contrast, omitting the morphology-aware alignment term (w/o sDTW) yielded marginal improvements in pointwise metrics, such as nRMSE and PCC, but substantially worsened the soft-DTW score. This divergent pattern indicates a fundamental trade-off between strict pointwise error minimization and morphology-aware temporal fidelity. This result suggests that soft-DTW supervision plays an important role in enforcing morphology-aware temporal alignment, thereby preventing the model from reducing amplitude errors at

the expense of clinically relevant temporal structure, such as interval durations and peak alignment.

Collectively, these ablation results support the view that both gated latent evolution and morphology-aware supervision are important for robust longitudinal waveform forecasting.

5. Discussion

The proposed CLEF provides a unified formulation for forecasting future ECG waveforms from heterogeneous historical records. By combining paper-to-digital signal recovery, shared latent representation learning, latent evolution modeling, and waveform-domain reconstruction, CLEF extends longitudinal ECG analysis beyond predefined scalar tracking toward waveform-level forecasting. Within the evaluated setting, this formulation enables year-scale prediction while preserving key waveform morphologies and diagnostically informative ECG structures.

The baseline comparison further clarifies the value of this formulation. Heuristic baselines, including persistence, averaging, and direct linear extrapolation, perform substantially worse than CLEF. This result indicates that future ECG morphology in this setting cannot be recovered through simple temporal continuation rules alone. Linear baselines improve over these heuristics but remain limited. This result suggests that low-capacity mappings in the signal domain are insufficient to model cross-visit electrophysiological change. Neural baselines further improve performance, yet still remain below CLEF across all metrics. Taken together, these results suggest that the advantage of CLEF does not arise from nonlinear prediction alone, but from future inference in a shared cross-temporal latent space with an explicit mechanism for latent evolution. The remaining gap between Gated Evo-MLP and CLEF further suggests that the performance gain is unlikely to be explained by gating alone, and is more plausibly associated with the full combination of shared representation learning, latent evolution modeling, and waveform-domain reconstruction.

The ablation results indicate that both the gated latent evolution module and morphology-aware waveform reconstruction supervision contribute meaningfully to performance. Replacing gated latent evolution with linear latent extrapolation leads to a consistent decline across metrics. This decline suggests that year-scale ECG change is not adequately captured by simple linear evolution in the shared latent space. Removing morphology-aware align-

ment slightly improves pointwise error metrics but substantially worsens soft-DTW. This divergent pattern indicates a trade-off between amplitude-wise reconstruction fidelity and morphology-aware temporal consistency. These findings support the view that gated latent evolution and morphology-aware supervision jointly contribute to robust future waveform inference.

This study also differs conceptually from prior waveform-oriented ECG modeling paradigms. Traditional longitudinal studies rely heavily on tracking predefined intervals, such as QTc or QRS duration, thereby compressing high-dimensional electrophysiological change into low-dimensional summaries [12, 7]. By contrast, most deep learning-based ECG studies remain focused on snapshot-based classification, anomaly detection, or static risk estimation, rather than explicit forecasting of future waveform morphology across serial visits [64]. By preserving a complete computational path from historical observations to future waveform reconstruction, CLEF provides a representation-learning formulation that supports cross-visit comparison and future waveform inference under heterogeneous archival conditions.

Despite these advances, several limitations should be acknowledged. First, this study was restricted to standardized 3-second Lead II segments aligned by the R peak. This controlled setting reduces the influence of respiration-induced baseline drift and rhythm variation. It allows a focused evaluation of heartbeat-level morphological progression. Accordingly, the reported findings should be interpreted as a foundational step toward heartbeat-level longitudinal morphology forecasting, rather than as a general solution for multi-lead or highly arrhythmic scenarios. Second, although the current latent evolution module captures gradual remodeling trajectories, it does not explicitly model abrupt waveform transitions triggered by intervening acute clinical events, such as sudden ischemic episodes or intermediate therapeutic interventions [65].

These limitations point to several directions for future research. Extending the observational scope from localized segments to full 12-lead recordings and longer Holter sequences represents an important next step. Methodologically, non-stationary dynamic models or decomposition-aware mechanisms may further improve the modeling of abrupt state transitions together with gradual electrophysiological remodeling.

Overall, the results support the view that longitudinal ECG evolution can be characterized more effectively in a shared latent space than through direct raw-signal comparison or predefined descriptor tracking alone. Within the evaluated setting, CLEF provides a practical formulation for individualized

trend characterization and waveform-level forecasting from historical ECG records.

6. Conclusion

This study presents CLEF, a unified framework for future ECG waveform forecasting from heterogeneous historical records. By combining paper-to-digital signal recovery, cross-temporal representation learning, latent evolution modeling, and waveform-domain reconstruction, CLEF enables year-scale forecasting while preserving key ECG waveform morphologies.

Experiments show that CLEF achieves an nRMSE of 0.077 and outperforms heuristic, linear, and neural baselines. This comparison shows that future ECG waveform forecasting requires more than simple persistence, direct extrapolation, or generic nonlinear prediction. Instead, effective forecasting requires a shared latent space that remains comparable across visits and supports subject-specific latent evolution.

These findings support the feasibility of extending longitudinal ECG analysis beyond predefined descriptor tracking toward waveform-level forecasting. Future work will examine whether this formulation generalizes to multi-lead recordings, longer observation windows, irregular follow-up intervals, and more heterogeneous clinical populations.

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